
ERRATUM TO "SLOPE BOUNDEDNESS AND EQUIDISTRIBUTION THEOREM"

by

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Introduction

We give a counterexample for the slope boundedness of an adelic line bundle even assuming its geometric part is big, which is provided by Xinyi Yuan via an argument by contradiction based on a consequence of the boundedness. Here we give the direct description.

1. The counterexample

Let E be an elliptic curve over a number field or a function field K , with a non-torsion line bundle $N \in \text{Pic}^0(E)$.

We may and do assume that $N = T_y^*L - L$ for some symmetric ample line bundle L on E and $y \in E(K) \setminus E_{\text{tors}}(K)$. Let $\bar{L}$ be a canonical adelic line bundle whose underlying line bundle is L . Set $\bar{N} = T_y^*\bar{L} - \bar{L}$.

Now consider $X = \mathbb{P}(T_y^*L \oplus L^\vee)$, and $M = \mathcal{O}_X(1)$. Note that M is big on X since the maximal slope of the vector bundle $T_y^*L \oplus L^\vee$ is $\deg_L(E) > 0$. Let

$$V_n = H^0(X, nM) = \bigoplus_{n_1+n_2=n} H^0(E, n_1T_y^*L - n_2L).$$

We equip V_n with the direct sum norms of the supnorms on $H^0(E, n_1T_y^*L - n_2L)$, and denote the adelic normed vector space by $\bar{V}_n$. Then

$$\hat{\mu}_{\min}(\bar{V}_n) = \min_{n_1+n_2=n} \{\hat{\mu}_{\min}(H^0(E, n_1T_y^*\bar{L} - n_2\bar{L}))\}.$$

If $\liminf_{n \rightarrow \infty} \frac{\widehat{\mu}_{\min}(\overline{V}_n)}{n} > -\infty$, then $\frac{1}{n_1+n_2} \widehat{\mu}_{\min}(H^0(E, n_1 T_y^* \overline{L} - n_2 \overline{L}))$ is bounded from below. In particular

$$\liminf_{n \rightarrow \infty} \frac{1}{(2m-1)n} \widehat{\mu}_{\min}(H^0(E, n(m(T_y^* \overline{L} - \overline{L}) + \overline{L})))$$

is bounded from below uniformly for $m \in \mathbb{N}_+$.

On the other hand, since $m(T_y^* L - L) + L = mN + L$ is ample,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \widehat{\mu}_{\min}(H^0(E, nmT_y^* \overline{L} - (nm - n)\overline{L})) = \inf_{x \in E(\overline{K})} h_{mT_y^* \overline{L} - m\overline{L} + \overline{L}}(x).$$

Notice that

$$\begin{aligned} h_{mT_y^* \overline{L} - m\overline{L} + \overline{L}}(x) &= mh_{\overline{L}}(x + y) - (m-1)h_{\overline{L}}(x) \\ &= m(h_{\overline{L}}(x) + h_{\overline{L}}(y) + 2\langle x, y \rangle) - (m-1)h_{\overline{L}}(x) \\ &= h_{\overline{L}}(x) + 2m\langle x, y \rangle + mh_{\overline{L}}(y) \\ &= h_{\overline{L}}(x + my) - (m^2 - m)h_{\overline{L}}(y). \end{aligned}$$

We have $h_{mT_y^* \overline{L} - m\overline{L} + \overline{L}}(-my) = -(m^2 + m)h_{\overline{L}}(y)$, which gives a contradiction as $\frac{-m^2+m}{2m-1}$ is unbounded from below.

So the asymptotic minimal slope in the sense of [1, 3, 4] is unbounded. Moreover, $\overline{N}$ can be taken to be relative semipositive, so the asymptotic minimal slope in the sense of [2] is also unbounded. $\square$

References

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